

The Projective Residue as a Schur Complement: Reduction of the Generation Split to the Chiral Asymmetry of Projection Locking

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Abstract

The fermionic sub-programme of Cosmochrony locates the three-generation mass split in the J_3 -odd part of the squared projective endomorphism E_Π^2 restricted to the gauge-singlet generation triplet C_{gen}^3 , parametrised by a single real number u through $E_\Pi^2|_{C_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2} + u, \frac{1}{2} - u)$, with the even sector $\text{diag}(1, \frac{1}{2}, \frac{1}{2})$ already closed by Born–Infeld parity [4]. This note fixes the structural status of u before any explicit construction of E_Π . First, the projected Dirac square admits a universal Feshbach/Schur form $E_\Pi = -\Pi_S D (1 - P) D \Pi_S^* = -M^\dagger M$ with $P = \Pi_S^* \Pi_S$, exhibiting E_Π as the Schur complement of the spinorial directions eliminated by the non-injective projection, negative semi-definite and vanishing in the injective limit. Second, the chiral block decomposition shows that u is controlled by the $\mathcal{D}^\pm$ -transported, generation-projected part of the antiunitary chiral equivariance defect

$$\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$$

of the eliminated block $1 - P$, rather than by a naive block difference. Third, the minimal non-injectivity $c \leftrightarrow q - c$ is chirally symmetric, so $u = 0$ at the level of axioms A1–A3; a non-zero u requires a chiral symmetry-breaking carried by the projection-locking axiom A4. The finite locking sector is proved J_Π -equivariant, so $u_{\text{fin}} = 0$, and chirality is shown to be a Lorentzian rather than a finite-fibre datum. Finally, the Schur-transversality branch required by the Born–Infeld genus companion is closed in the present Lorentzian spin stratum: the projected A4 commutator has the exact metaplectic opening

$$\alpha(t, s) = ts \frac{r}{\sinh r} = ts + O((ts)^2), \quad \mu(t, s) \equiv 0,$$

with $\cosh r = 1 + \frac{ts}{2}$, and its timelike Clifford symbol is not in the transported zero-mode locus. Together with the electric-genus result of the A4 companion, this promotes the existence statement to $u \neq 0$ in the present stratum. The remaining open deliverables are the explicit Lorentzian eliminated block $1 - P(s)$, the absolute normalisation $|u|$, and the projected Yukawa sector.

Keywords: projective endomorphism; Schur complement; non-injective projection; chiral asymmetry; projection locking; Born–Infeld saturation; generation split; Seeley–DeWitt expansion; admissible spinor bundle.

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1 Position and purpose

The fermionic sub-programme [6] derives the admissible spinor bundle $\mathcal{S}_\Pi$, the $V - A$ chiral structure, hypercharge rigidity, and a gauge-singlet three-generation factor C_{gen}^3 as the spinorial face of the admissible Weil module after the Lorentzian complexification of its metaplectic lift. The generation mass split is carried by the projective endomorphism $E_\Pi \in \Gamma(\text{End}(\mathcal{S}_\Pi))$, the zero-order spectral residue of non-injective projection in the spinorial sector [6]. On the rank-three admissible triplet $C_{\text{gen}}^3 = \text{span}(e_0, e_+, e_-)$ with Cartan generator $J_3 = \text{diag}(0, 1, -1)$, the squared residue takes the normal form

$$E_\Pi^2|_{C_{\text{gen}}^3} = \frac{C_2 - J_3^2}{C_2} + u J_3 + v R_{\text{mix}}, \quad (1)$$

where $C_2 = 2$ is the spin-one Casimir, R_{mix} is the off-diagonal generation mixing, and u is the single real coefficient governing the split. The CP-even/real sector imposes $v = 0$, and the even part is closed analytically by Born–Infeld parity on the equatorial admissible fibre [4]:

$$(E_\Pi^2)_{\text{even}}|_{C_{\text{gen}}^3} = \frac{C_2 - J_3^2}{C_2} = \text{diag}(1, \tfrac{1}{2}, \tfrac{1}{2}). \quad (2)$$

The only remaining unknown is therefore

$$u = \frac{\langle \Pi_{J_\Pi\text{-odd}}(E_\Pi^2), J_3 \rangle_{\text{HS}}}{\langle J_3, J_3 \rangle_{\text{HS}}} = \tfrac{1}{2} (\langle e_+ | E_\Pi^2 | e_+ \rangle - \langle e_- | E_\Pi^2 | e_- \rangle). \quad (3)$$

This note records what is now established about u and isolates the single conceptual obstruction that remains, so that the explicit construction of E_Π can proceed in a locked frame. The companion front observables of the angular and oriented-frontier notes [4] are not the locus of u ; they explain the obstructions and provide null controls, but u lives in the zero-order residue of the projected Dirac operator. The numerical front controls are recorded in Appendix A.

2 The projective residue as a Schur complement

Let $\Pi_S : \mathcal{S} \rightarrow \mathcal{S}_\Pi$ be the restriction of the admissible projection to the spinorial bundle, Π_S^* the admissible minimal lift, and

$$D_{\Pi,g,A} := \Pi_S \circ D_{g,A} \circ \Pi_S^*, \quad P := \Pi_S^* \Pi_S, \quad (4)$$

following [6]. One does not assume $\Pi_S \Pi_S^* = \text{id}_{\mathcal{S}_\Pi}$; this holds only in the injectively resolved limit. Symbol-compatibility means the projection defect affects only the zero-order part of $D_{\Pi,g,A}^2$, leaving the Clifford principal symbol intact.

Proposition 1 (Schur form of the projective residue). *Under symbol-compatibility, the projective endomorphism of [6] satisfies*

$$E_\Pi = -\Pi_S D_{g,A} (1 - P) D_{g,A} \Pi_S^* = -M^\dagger M, \quad M := (1 - P) D_{g,A} \Pi_S^*. \quad (5)$$

In particular $E_\Pi \preceq 0$, it is zero-order, and $E_\Pi = 0$ if and only if $1 - P = 0$, i.e. in the injective limit.

Proof. With $P = \Pi_S^* \Pi_S$ idempotent, $D_{\Pi,g,A}^2 = \Pi_S D_{g,A} P D_{g,A} \Pi_S^*$. The uncompressed square $\Pi_S D_{g,A}^2 \Pi_S^*$ reconstructs the effective spin–gauge Lichnerowicz operator on $\mathcal{S}_\Pi$ up to symbol, so that

$$\text{Lich}(g, A) \approx \Pi_S D_{g,A}^2 \Pi_S^* = D_{\Pi,g,A}^2 + \Pi_S D_{g,A} (1 - P) D_{g,A} \Pi_S^*.$$

Subtracting the Lichnerowicz block from $D_{\Pi,g,A}^2$ gives (5). Since $1 - P$ is a self-adjoint projector and $D_{g,A}$ is self-adjoint, $\Pi_S D_{g,A}(1 - P)D_{g,A}\Pi_S^* = M^\dagger M \succeq 0$, hence $E_\Pi = -M^\dagger M \preceq 0$. The first-order part vanishes by symbol-compatibility, leaving a zero-order endomorphism, and $E_\Pi = 0 \Leftrightarrow M = 0 \Leftrightarrow (1 - P)D_{g,A}\Pi_S^* = 0$, which for a faithful symbol holds exactly when $1 - P = 0$. $\square$

Proposition 1 makes precise the statement of [6] that E_Π is not a postulated finite Dirac datum but is induced by the non-injective projection itself: it is the Schur complement of the eliminated sector $1 - P$.

3 Chiral block reduction and the localisation of u

The chirality operator $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ of the effective Clifford algebra exists as a global section because of the Lorentzian closure of the effective metric [9]. The Dirac operator is chirality-odd, hence off-diagonal in the chiral splitting $\mathcal{S}_\Pi = \mathcal{S}_L \oplus \mathcal{S}_R$:

$$D_{g,A} = \begin{pmatrix} 0 & \mathcal{D}^- \\ \mathcal{D}^+ & 0 \end{pmatrix}, \quad 1 - P = \begin{pmatrix} \pi_{LL} & \pi_{LR} \\ \pi_{RL} & \pi_{RR} \end{pmatrix}, \quad (6)$$

with π_{LL}, π_{RR} self-adjoint and $\pi_{RL} = \pi_{LR}^\dagger$.

Proposition 2 (Chiral diagonal blocks of the residue). *With the conventions of (6),*

$$E_{LL} = -\Pi_S \mathcal{D}^- \pi_{RR} \mathcal{D}^+ \Pi_S^*, \quad E_{RR} = -\Pi_S \mathcal{D}^+ \pi_{LL} \mathcal{D}^- \Pi_S^*. \quad (7)$$

The left-admissibility condition $P_R E_\Pi P_R = 0$ of [6] is equivalent to $\pi_{LL} = 0$: the non-injective projection eliminates no purely left-handed direction.

Proof. Direct multiplication of (6) gives

$$D_{g,A}(1 - P)D_{g,A} = \begin{pmatrix} \mathcal{D}^- \pi_{RR} \mathcal{D}^+ & \mathcal{D}^- \pi_{RL} \mathcal{D}^- \\ \mathcal{D}^+ \pi_{LR} \mathcal{D}^+ & \mathcal{D}^+ \pi_{LL} \mathcal{D}^- \end{pmatrix}.$$

Conjugating by Π_S, Π_S^* and using $E_\Pi = -\Pi_S D_{g,A}(1 - P)D_{g,A}\Pi_S^*$ gives (7). The right-right block of E_Π is $E_{RR} = -\Pi_S \mathcal{D}^+ \pi_{LL} \mathcal{D}^- \Pi_S^*$, which vanishes for a faithful Weyl symbol exactly when $\pi_{LL} = 0$. $\square$

Corollary 1 (Localisation of u). *After restriction to C_{gen}^3 , the J_3 -odd part of E_Π^2 is controlled by the chiral asymmetry of the eliminated block,*

$$u \propto \pi_{RR} - \pi_{LL}, \quad (8)$$

dressed by the Weyl transport $\mathcal{D}^\pm$. The split u is therefore an intrinsic spectral asymmetry of the chiral part of the projective residue, not a shell average of any front cocycle.

This raw block expression is refined in Section 5, where the antiunitary nature of J_Π replaces the naive block difference by the transported defect $\Delta_\chi(P)$.

4 Stratification: u vanishes unless A4 breaks chiral symmetry

The minimal non-injectivity is the parity involution $c \leftrightarrow q - c$ on conjugate Weil blocks [8]. Its spinorial lift is the antilinear Born–Infeld parity J_Π , which exchanges the chiral subbundles, $J_\Pi P_L = P_R J_\Pi$ [6]. A parity that exchanges $\mathcal{S}_L \leftrightarrow \mathcal{S}_R$ treats the two chiralities symmetrically, so the minimal eliminated block is chirally symmetric, $\pi_{LL} = \pi_{RR}$.

Proposition 3 (Chiral-symmetry stratification of u). *If $\pi_{LL} = \pi_{RR}$ then $u = 0$. The conjugate pair $\{V^{(c)}, V^{(q-c)}\}$ carries no preferred orientation at the level of axioms A1–A3, consistently with the non-premature-selection axiom A3 [1]. Hence at the A1–A3 level $u = 0$. A non-zero u requires $\pi_{LL} \neq \pi_{RR}$, a chiral symmetry-breaking that can originate only in the projection-locking axiom A4.*

Proof. By Corollary 1, $u \propto \pi_{RR} - \pi_{LL}$, so $\pi_{LL} = \pi_{RR}$ forces $u = 0$. The minimal non-injectivity is the chirally symmetric parity $c \leftrightarrow q - c$, and A3 forbids selection among open directions before locking, so no chiral orientation is fixed by A1–A3. The only axiom that selects a resolved direction is A4, “resolution occurs only when continuous Born–Infeld saturation meets the discrete shell support” [1]; any chiral asymmetry $\pi_{LL} \neq \pi_{RR}$ must therefore arise there. $\square$

The left-admissibility branch $\pi_{LL} = 0$ [6] is thus the maximal chiral asymmetry compatible with A4 locking, and the sign of u is the spontaneous $V - A$ choice exchanged under $P_L \leftrightarrow P_R$. The even sector (2) is the chirally symmetric part of the same Born–Infeld saturation [4, 2]; u is the chirally asymmetric part of that same locking. This is the structural sense in which the Higgs amplitude mode of the saturated projective condensate and the generation split are read from a single A4 event.

5 Spinorial locking and the J_Π -equivariance defect

This section refines Proposition 3: the chirally symmetric branch is the transported equivalence $\pi_{LL} = \tau \overline{\pi_{RR}} \tau^{-1}$, not the naive block equality $\pi_{LL} = \pi_{RR}$, the bar being the complex conjugation carried by the antiunitary Born–Infeld lift J_Π .

Let $V = \text{span}(v_+, v_-)$ denote the fundamental spinorial doublet, with Cartan generator $Hv_\pm = \pm v_\pm$. The spinorial Born–Infeld lift of [6] is the antiunitary structure

$$J_\Pi = \epsilon \circ \text{conj}, \quad \epsilon = \sigma_z \sigma_x = i \sigma_y, \quad (9)$$

so that, in the standard convention, $J_\Pi v_+ = -v_-$ and $J_\Pi v_- = v_+$. Thus $J_\Pi^2 = -1$ on V : this is the pseudoreal, or quaternionic, structure of the fundamental spinorial sector.

On the symmetric square $\text{Sym}^2(V) = \text{span}(e_0, e_+, e_-)$ with $e_0 = \sqrt{2} v_+ v_-$, $e_+ = v_+^2$, $e_- = v_-^2$, the induced antiunitary $J_\Pi^{(2)}$ acts by

$$J_\Pi^{(2)} e_0 = -e_0, \quad J_\Pi^{(2)} e_+ = e_-, \quad J_\Pi^{(2)} e_- = e_+. \quad (10)$$

Consequently $(J_\Pi^{(2)})^2 = +1$ on $\text{Sym}^2(V)$: the generation triplet carries a real involution, although the underlying doublet is pseudoreal. With $J_3 = \text{diag}(0, 1, -1)$ on (e_0, e_+, e_-) , the formulas (10) give

$$J_\Pi^{(2)} J_3 (J_\Pi^{(2)})^{-1} = -J_3, \quad (11)$$

so the Born–Infeld spinorial conjugation exchanges the two outer generation weights.

We now apply this to the Schur residue. Let $\tau : \mathcal{S}_R \rightarrow \mathcal{S}_L$ be the unitary intertwiner realising the chirality exchange $J_\Pi P_L = P_R J_\Pi$ of [6] in the chiral frame. Equivariance of the locking projector, $[P, J_\Pi] = 0$, then forces on the diagonal chiral blocks of $1 - P$ not the naive equality but the transported, complex-conjugated equivalence

$$\pi_{LL} = \tau \overline{\pi_{RR}} \tau^{-1}. \quad (12)$$

The relevant obstruction is consequently the antiunitary defect

$$\Delta_\chi(P) := \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}, \quad (13)$$

which reduces to $\pi_{LL} - \tau \pi_{RR} \tau^{-1}$ on the real structure fixed by $J_\Pi^{(2)}$ on the generation triplet.

Lemma 1 (Chiral obstruction of spinorial locking). *Assume the projective Dirac transport is J_Π -compatible, $J_\Pi D_{g,A} J_\Pi^{-1} = \pm D_{g,A}$ and $J_\Pi \Pi_S J_\Pi^{-1} = \Pi_S$, as in the spinorial Born–Infeld lift of [6]. If the spinorial locking projector is J_Π -equivariant, $[P, J_\Pi] = 0$, then the Schur residue $E_\Pi = -\Pi_S D_{g,A} (1 - P) D_{g,A} \Pi_S^*$ satisfies $J_\Pi E_\Pi^2 J_\Pi^{-1} = E_\Pi^2$. Consequently $\langle e_+ | E_\Pi^2 | e_+ \rangle = \langle e_- | E_\Pi^2 | e_- \rangle$, and the split coefficient u of (3) vanishes. Hence a non-zero generation split requires a failure of J_Π -equivariance at projection locking,*

$$u \neq 0 \implies [P, J_\Pi] \neq 0.$$

Proof. Since $J_\Pi^{(2)} e_+ = e_-$ and (11) holds, the two outer generation weights are conjugate under the Born–Infeld spinorial involution. The assumptions $J_\Pi D_{g,A} J_\Pi^{-1} = \pm D_{g,A}$, $J_\Pi \Pi_S J_\Pi^{-1} = \Pi_S$ and $[P, J_\Pi] = 0$ imply $J_\Pi E_\Pi J_\Pi^{-1} = \pm E_\Pi$, up to the irrelevant sign induced by the chirality-odd Dirac transport. Squaring removes this sign and gives $J_\Pi E_\Pi^2 J_\Pi^{-1} = E_\Pi^2$. Because J_Π is antiunitary, $\langle J_\Pi \psi | E_\Pi^2 | J_\Pi \psi \rangle = \overline{\langle \psi | E_\Pi^2 | \psi \rangle}$, and E_Π^2 is self-adjoint, so this expectation value is real. Taking $\psi = e_+$ and using $J_\Pi e_+ = e_-$ gives $\langle e_- | E_\Pi^2 | e_- \rangle = \langle e_+ | E_\Pi^2 | e_+ \rangle$. The definition (3) of u then gives $u = 0$, and the final implication is the contrapositive. $\square$

Remark 1 (Necessary condition, not a sufficient one). The condition $[P, J_\Pi] \neq 0$ is necessary for $u \neq 0$, but not sufficient. The commutator may be supported outside the generation triplet, or in components that do not survive the Dirac transport and the final projection to C_{gen}^3 . The magnitude of u is therefore not determined by the abstract commutator itself: it is set by the $\mathcal{D}^\pm$ -transported, generation-projected part of the antiunitary defect $\Delta_\chi(P)$ of (13). Determining this transported and projected component is the remaining A4-level deliverable.

6 Locked Seeley–DeWitt conventions: operator u equals spectral u

The projected square is Laplace-type, $D_{\Pi,g,A}^2 = -(\nabla^{S,A})^2 + E$, with $E = \frac{R}{4} + \frac{1}{2} \gamma^\mu \gamma^\nu F_{\mu\nu}^a T^a + E_\Pi$ [6], and the γ_5 -weighted Seeley–DeWitt coefficient $a_4 = (4\pi)^{-2} \text{tr}(\frac{1}{2} E^2 - \frac{1}{6} R E + \dots)$ in the convention of [6] consistent with the heat-kernel manual. The bosonic synthesis fixes the spectral normalisation: a_2 is the horizontal gravitational response and a_4 the vertical gauge response of the spectral entropy functional, with induced couplings carrying a common factor at each Seeley–DeWitt order [7, 13].

Three facts make the operator-level u of (3) coincide with the spectral coefficient that enters the effective amplitude. First, u is quadratic in E_Π , so it is immune to the linear sign convention of E . Second, on the gauge-singlet C_{gen}^3 the gauge curvature term vanishes, and on the flat effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ [9] all curvature cross-terms vanish, in particular $-\frac{1}{6} R E_\Pi = 0$; only $\frac{1}{2} E_\Pi^2$ survives, so no linear-in- E_Π term can separate the operator and spectral readings. Third, the common a_4 normalisation factor cancels in the generation ratio $(\frac{1}{2} + u)/(\frac{1}{2} - u)$, so u is independent of the spectral normalisation while the absolute mass scale carries it. A curvature correction $-\frac{1}{6} R E_\Pi$ would reappear only on a curved background, as a gravitational correction to masses, and does not affect the structural u .

7 Chirality is not a finite-fibre label

The chirality γ_5 is the chirality operator of the effective spacetime Clifford algebra [6], existing because of the Lorentzian closure $g^{\mu\nu} = 2\eta^{\mu\nu}$ [9]. The finite Weil module selects only the compact pseudoreal form $\mathfrak{su}(2)$, in which the fundamental is self-conjugate [3]; the Lorentzian complexification is what removes the equivalence and renders $\mathcal{S}_L$ and $\mathcal{S}_R$ inequivalent [6].

Remark 2 (No finite chiral edge label). A cascade edge $g \rightarrow gs$ is a Heisenberg translation, not a symplectic element, so it carries no canonical metaplectic lift; and the chiral eigendirections are the complex combinations of the real generators, so a real edge has no definite chirality. On the admissible front the only orientation-odd coordinate datum reduces to the sign of the cocycle, which is the eliminated-amplitude proxy, not a chirality. Consequently u cannot be extracted from any finite front observable: the chiral doubling $\mathcal{S}_L \oplus \mathcal{S}_R$ and the zero-order Dirac residue are required. This explains, structurally, every vanishing of the symmetric front and shell observables: each lacks the chiral asymmetry that lives in the Lorentzian residue.

8 Finite locking is J_Π -equivariant: separation of the finite and Lorentzian sectors

Before any Lorentzian computation, the finite sector must be excluded as a source of u . This section proves that the finite projection locking is J_Π -equivariant, so that u cannot originate in the finite cascade and the generation split is forced to be a Lorentzian datum.

Let $\text{Lock}_S^{\text{fin}}$ denote the projection-locking data of axiom A4 [1] at finite q : the continuous Born–Infeld saturation condition intersected with the discrete shell support on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Write $P_{\text{fin}} = \Pi_S^* \Pi_S$ for the resulting finite admissible projector.

Proposition 4 (Finite equivariance of projection locking). *For q an odd prime, the finite locking data is invariant under the spinorial Born–Infeld parity J_Π , and the finite admissible configuration space carries no J_Π -odd modulus. Consequently*

$$[\text{Lock}_S^{\text{fin}}, J_\Pi] = 0, \quad [P_{\text{fin}}, J_\Pi] = 0, \quad (14)$$

and by Lemma 1 the finite generation split vanishes, $u_{\text{fin}} = 0$.

Proof. Both constituents of A4 are J_Π -invariant at finite q . The continuous saturation condition is Born–Infeld parity itself, whose spinorial lift is J_Π [6], hence J_Π -invariant by construction. The discrete shell support is a union of orbits of the parity $c \leftrightarrow q - c$ [8]; for q an odd prime this involution acts as $c \mapsto -c$ on the central character, whose only fixed point is $c = 0$, so no non-trivial shell is parity-fixed and the support singles out no orientation. The locking data is therefore J_Π -invariant. Equivariance of P_{fin} then follows from the absence of a J_Π -odd deformation: a cascade edge $g \rightarrow gs$ is a Heisenberg translation, not a symplectic element, so it carries no canonical metaplectic lift, and the chiral eigendirections are complex combinations created only by Lorentzian complexification [6]. There is thus no finite direction along which a J_Π -conjugate pair of locked branches could split. The finite locked branch is therefore J_Π -equivariant, giving (14). The vanishing $u_{\text{fin}} = 0$ is then Lemma 1. $\square$

The numerical front controls of Appendix A confirm (14) directly, though they do not prove it: the proof is the absence of a finite odd modulus above. The symmetric front average vanishes identically, $\Theta_{\text{raw}} = 0$ for $q \in \{61, 101, 151\}$, and the oriented frontier is re-paired by the residual automorphism $\varphi = R_b \circ J_\Pi$, the finite avatar of the parity. These zeros are not failures to detect u ; they are the measured signature of finite J_Π -equivariance.

Corollary 2 (Finite/Lorentzian separation). *The finite A4 locking cannot generate u . A non-zero generation split can arise only in the Lorentzian completion of the Born–Infeld saturation, where the chiral-odd modulus first exists.*

This separates the remaining problem cleanly. Let s parametrise the J_Π -odd deformation of the locked block, opened only by Lorentzian complexification, with $J_\Pi : s \mapsto -s$, so that $\Delta_\chi(P)$ and hence u are odd in s . Since J_Π is a symmetry of the Lorentzian saturation functional $\mathcal{B}$,

one has $\mathcal{B}(s) = \mathcal{B}(-s)$, so $s = 0$ is automatically stationary and the locking character is fixed by the chiral curvature

$$\mu_\chi^2 := \partial_s^2 \mathcal{B}(0). \quad (15)$$

Interpreting A4 as selection of saturated admissible minima of $\mathcal{B}$,

$$\mu_\chi^2 > 0 \Rightarrow J_\Pi\text{-equivariant locking, } u = 0; \quad \mu_\chi^2 < 0 \Rightarrow \text{a } J_\Pi\text{-conjugate saturated pair } \pm s_*, u \neq 0,$$

and the marginal case $\mu_\chi^2 = 0$ is decided by the leading non-vanishing even coefficient λ_{2k} of $\mathcal{B}(s) = \mathcal{B}(0) + \lambda_{2k}s^{2k} + O(s^{2k+2})$: positive stabilises the symmetric point ($u = 0$), negative drives a higher-order spontaneous break ($u \neq 0$), and vanishing to all accessible orders leaves u an unfixed modulus, signalling that A4 alone does not determine the split. This marginal branch is a degenerate case of the second variation, not an independent source, so the decisive step is the necessarily Lorentzian computation of μ_χ^2 , carried out in the A4 companion [10]: in the Schur-transverse branch the chiral polarisation is electric, so $\mu_\chi^2 < 0$ and the split opens spontaneously.

This does not exclude a chiral frontier normalisation after Lorentzian complexification. It excludes only a symmetric finite-front observable. Once γ_5 exists, a J_Π -odd frontier cocycle may supply the discrete factor entering the A4 matching [12], while the existence and stability of the split remain governed by the Lorentzian saturation curvature μ_χ^2 . This discrete factor is identified in [5]: the chiral weight is the Weyl representation type, and in the inherited Heisenberg central-phase lift the frontier normalisation is fixed maximally; the split $u \neq 0$ nonetheless still requires the Schur transversality below, on the same chiral object.

9 Schur transversality of the A4 locking defect

Remark 1 left the A4-level deliverable as a necessary condition that is not yet a criterion: the commutator $[P, J_\Pi] \neq 0$ may be supported outside the generation triplet, or may die under the Weyl transport and the projection to C_{gen}^3 . Section 8 has since reduced the modulus to the Lorentzian deformation s with saturation curvature μ_χ^2 (15). We now turn the necessary condition into an exhaustive dichotomy on the transported, generation-projected defect.

With s the Lorentzian J_Π -odd deformation parameter of Section 8, differentiating the Schur form (5) at $s = 0$ with $D_{g,A}$ fixed, and using $\partial_s[-(1 - P)] = \partial_s P$, gives the first-order chiral polarisation

$$\mathcal{P} := \partial_s E_\Pi|_0 = \Pi_S D_{g,A} (\partial_s P|_0) D_{g,A} \Pi_S^*. \quad (16)$$

The positive sign is the derivative of $-(1 - P)$, not a convention. The induced first-order contribution to E_Π^2 is the anticommutator $\{E_\Pi|_0, \mathcal{P}\}$ with the J_Π -even background; since u is odd in s , its leading term is linear, so the vanishing of (16) on the outer weights removes the first-order opening of the split, leaving the marginal branch of (15).

Lemma 2 (Weyl-transport faithfulness on non-zero external symbols). *Let X be supported on the outer generation weights $\text{span}(e_+, e_-)$. The transported Weyl-symbol map $X \mapsto \Pi_S D_{g,A} X D_{g,A} \Pi_S^*$ inherits the non-degeneracy of the Clifford principal symbol already used in Proposition 2. It can annihilate a non-zero external symbol only when that symbol lies in the null space of the transported map, equivalently when its source or target component lies in the corresponding zero-mode sector of $\mathcal{D}^\pm$. This is the symbol-level faithfulness invoked in the proof of Proposition 2; it does not assert the absence of zero modes, whose locus is classified in Proposition 5 as a Schur-null branch.*

Definition 1 (Schur transversality). The A4 locking defect is *Schur-transverse* on the generation triplet if the chiral polarisation (16) does not vanish on the outer weights,

$$\Pi_S D_{g,A} (\partial_s P|_0) D_{g,A} \Pi_S^*|_{C_{\text{gen}}^3, J_\Pi\text{-odd}} \neq 0. \quad (17)$$

Otherwise it is *Schur-null*.

Proposition 5 (Schur dichotomy for the A4 locking defect). *Let σ denote the projection of $[\partial_s P|_0, J_\Pi]$ onto $\text{span}(e_+, e_-)$. On the chiral-odd A4 locus exactly one of the following alternatives holds.*

1. *If $\sigma = 0$, the opened direction carries no J_Π -odd component on the outer generation weights, the polarisation (17) vanishes, and the locking is Schur-null.*
2. *If $\sigma \neq 0$ but σ lies in the transported zero-mode locus of $\mathcal{D}^\pm$, the polarisation (17) again vanishes and the locking is Schur-null.*
3. *If $\sigma \neq 0$ and σ does not lie in the transported zero-mode locus, the polarisation (17) is non-zero and the locking is Schur-transverse.*

In the Schur-null cases the first-order Schur polarisation on the outer weights is absent. The split does not open at first order and the saturation curvature μ_χ^2 is marginal, the locking falling to the marginal branch of (15).

Proof. By (10)–(11) the antiunitary $J_\Pi^{(2)}$ exchanges $e_+ \leftrightarrow e_-$, so the J_Π -odd component of $\partial_s P|_0$ on the outer weights is exactly the projection σ of $[\partial_s P|_0, J_\Pi]$ onto $\text{span}(e_+, e_-)$. If $\sigma = 0$ the input to the transport (16) has no surviving external-weight component, so the image vanishes; this implication uses only that the projection to C_{gen}^3 retains the $e_\pm$ weights, not Weyl-transport faithfulness. If $\sigma \neq 0$ but lies in the transported zero-mode locus, the map of Lemma 2 sends it to zero. In both cases (17) vanishes, the first-order polarisation is absent, and the split is undecided at leading order, falling to the marginal branch. Conversely, if $\sigma \neq 0$ and does not lie in the zero-mode locus, Lemma 2 gives a non-zero transported image, so (17) holds. Faithfulness is used only in this converse direction, to prevent a non-zero external J_Π -odd symbol from dying under the Schur transport; the Schur-null implication does not use it. $\square$

Lemma 3 (Exact metaplectic opening of the Schur defect). *Let $V \simeq \mathbb{C}^2$ carry the standard $\mathfrak{sl}_2$ -triple*

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = [E, F],$$

with $H = 2J_3$ on the outer generation weights of $\text{Sym}^2(V) = \langle e_0, e_+, e_- \rangle$, and let $g(t, s) = \exp(tE)\exp(sF)$ be the oriented metaplectic step of the cascade bridge [6]. Then

$$g(t, s) = \begin{pmatrix} 1 + ts & t \\ s & 1 \end{pmatrix}, \quad \det g(t, s) = 1,$$

and, with $\cosh r = 1 + \frac{ts}{2}$, in the non-parabolic regular domain

$$\log g(t, s) = \frac{r}{\sinh r} \left(tE + sF + \frac{ts}{2}H \right).$$

Consequently the J_Π -odd Cartan component transported to the outer generation doublet is

$$\alpha(t, s) = ts \frac{r}{\sinh r} = ts + O((ts)^2),$$

while the R_{mix} component vanishes identically, $\mu(t, s) \equiv 0$. In particular $\alpha(t, s) \not\equiv 0$, and $\alpha(t, s) \neq 0$ for every non-degenerate oriented step $ts \neq 0$ outside the removable logarithmic singularities.

Proof. The matrix form follows from $E^2 = F^2 = 0$, giving $\exp(tE) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ and $\exp(sF) = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}$. The element $g(t, s)$ lies in SL_2 with $\text{tr } g(t, s) = 2 + ts$, hence $\cosh r = \frac{1}{2} \text{tr } g(t, s) = 1 + \frac{ts}{2}$. For a non-parabolic SL_2 element the logarithm is $\log g = \frac{r}{\sinh r} (g - \frac{1}{2}(\text{tr } g)\mathbb{K})$, and here $g(t, s) - \frac{1}{2}(\text{tr } g)\mathbb{K} = tE + sF + \frac{ts}{2}H$. Since $H = 2J_3$, the J_3 coefficient is $ts(r/\sinh r)$, and

$r/\sinh r = 1 + O(ts)$ gives $\alpha(t, s) = ts + O((ts)^2)$. Lifting to $\text{Sym}^2(V)$, the horizontal generators E, F send the outer weights $e_{\pm}$ to the central weight e_0 , so their restriction to $\langle e_+, e_- \rangle$ carries no R_{mix} component; only the Cartan J_3 survives, whence $\mu(t, s) \equiv 0$, the mixing direction being absent in the real symplectic model [6]. The factor $r/\sinh r$ is strictly positive on the regular real domain, so $\alpha(t, s) = 0$ only at the degenerate step $ts = 0$. $\square$

Corollary 3 (Schur transversality in the present spin stratum). *Assume the symbol-compatibility of the projected Dirac operator [6] and the Lorentzian identification of the central commutator $Z = [X, Y] \simeq \partial_\tau$ of the effective metric closure [9]. Then the A_4 locking defect is Schur-transverse on the generation triplet in the present Lorentzian spin stratum,*

$$\Pi_S D_{g,A} (\partial_s P|_0) D_{g,A} \Pi_S^*|_{C_{\text{gen}}^3, J_{\Pi\text{-odd}}} \neq 0.$$

Proof. By Lemma 3 the transported defect contains a non-zero $J_{\Pi\text{-odd}}$ Cartan component $\sigma = \alpha(t, s) J_3$ for every non-degenerate oriented step, so the Schur-null case $\sigma = 0$ is excluded. Its source is the commutator $Z = [X, Y]$, identified with the timelike direction ∂_τ in the Lorentzian effective metric, and the Clifford principal symbol is non-degenerate on non-zero timelike covectors. Therefore the transported J_3 -symbol is not in the zero-mode locus of the Weyl blocks $\mathcal{D}^\pm$, so the transported zero-mode case is excluded. The remaining alternative of Proposition 5 is the Schur-transverse branch. $\square$

Corollary 4 (A_4 -level reduction). *The A_4 residual problem is reduced to the projected Lorentzian datum*

$$\sigma = [\partial_s P|_0, J_{\Pi}]|_{\text{span}(e_+, e_-)}.$$

If the locking is Schur-transverse, the generation split is decided by the Born–Infeld genus of the chiral polarisation [10]: $\mu_\chi^2 > 0$ gives $u = 0$, and $\mu_\chi^2 < 0$ gives a spontaneous $V - A$ split $u \neq 0$. If the locking is Schur-null, $\mu_\chi^2 = 0$ and the split is governed by the leading even coefficient λ_{2k} of (15).

Schur transversality is thereby closed in the present Lorentzian spin stratum: Lemma 3 evaluates the projected A_4 commutator in closed form and excludes the exact-zero branch, and Corollary 3 excludes the transported zero-mode branch, so the dichotomy of Proposition 5 selects the Schur-transverse case. The Lorentzian datum $\partial_s P|_0$ is the metaplectic opening $g(t, s) = \exp(tE) \exp(sF)$, fixed together with the genus of [10] by the Born–Infeld saturation structure [2].

10 Status of results

Proved / structural (this note).

- $E_{\Pi} = -\Pi_S D(1 - P) D \Pi_S^* = -M^\dagger M$ is the Schur complement of the eliminated sector, negative semi-definite, zero-order, vanishing in the injective limit (Proposition 1).
- Chiral diagonal blocks (7); left-admissibility $\Leftrightarrow \pi_{LL} = 0$ (Proposition 2).
- The J_3 -odd component of E_{Π}^2 is first exposed in the chiral block decomposition as a raw asymmetry $\pi_{RR} - \pi_{LL}$ (Corollary 1), and refined in Section 5 as the antiunitary equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$, after $\mathcal{D}^\pm$ -transport and projection to C_{gen}^3 .
- $\pi_{LL} = \pi_{RR} \Rightarrow u = 0$; non-zero u requires A_4 chiral symmetry-breaking (Proposition 3).
- The symmetric-square action of the Born–Infeld involution is fixed: $J_{\Pi}^2 = -1$ on the fundamental doublet, $(J_{\Pi}^{(2)})^2 = +1$ on the generation triplet, and it anticommutes with

J_3 (11). Hence $u \neq 0$ requires a failure of J_Π -equivariance at projection locking (Lemma 1); the magnitude of u depends on the $\mathcal{D}^\pm$ -transported, generation-projected component of the antiunitary defect $\Delta_\chi(P)$ (13).

- Operator u equals spectral u on the flat effective metric; u normalisation-independent in the ratio (Section 6).
- Chirality is Lorentzian, not a finite-fibre label; u is not a front observable (Remark 2).
- Finite projection locking is J_Π -equivariant, so $u_{\text{fin}} = 0$: u cannot originate in the finite cascade and is a Lorentzian datum (Proposition 4, Corollary 2).
- Schur transversality is proved in the present Lorentzian spin stratum: Lemma 3 excludes the exact-zero branch through the closed form $\alpha(t, s) = ts(r/\sinh r) \neq 0$ of the projected A4 commutator, and Corollary 3 excludes the transported zero-mode branch under symbol-compatibility and the timelike-source identification $Z = [X, Y] \simeq \partial_\tau$. Proposition 5 then selects the Schur-transverse branch.

Conditional. The left-admissibility branch $\pi_{LL} = 0$ [6] is an admissible branch selected by the orientation-compatible assumption; its derivation from A4 is open.

Open. The existence of the split is closed in the present Lorentzian spin stratum after the Schur-transversality closure of Section 9 and the electric-genus result of the A4 companion. What remains open is not $u \neq 0$, but the explicit Lorentzian eliminated block $1 - P(s)$, the absolute normalisation $|u|$, and the projected Yukawa sector. The magnitude $|u|$ depends on the fully transported and projected Born–Infeld saturation data, including the chiral-frontier normalisation, and is not determined by the Schur-transversality argument itself.

11 Open deliverables

The finite contribution to A4 is now separated: finite projection locking is J_Π -equivariant and cannot generate u (Proposition 4 and Corollary 2). The Schur-transversality test that previously conditioned the A4 companion is closed in the present Lorentzian spin stratum: by Lemma 3 and Corollary 3 the A4 direction is Schur-transverse. The Born–Infeld genus of [10] therefore selects the electric branch, $\mu_\chi^2 < 0$, and the generation split opens spontaneously, $u \neq 0$.

The remaining open deliverables are no longer the existence of the Schur-transverse branch, nor the sign of the Lorentzian genus. They are:

1. the explicit Lorentzian eliminated block $1 - P(s)$ entering the Schur complement

$$E_\Pi = -\Pi_S D_{g,A} (1 - P(s)) D_{g,A} \Pi_S^*;$$

2. the absolute normalisation $|u|$, including the fully transported Born–Infeld saturation data, the cubic Schur backreaction, and the chiral-frontier normalisation;
3. the projected Yukawa sector and the corresponding mass matrices.

The structural form of the first deliverable is fixed in the companion note [11]: the eliminated-block jet $1 - P(s) = Q_0 + sQ_1 + s^2Q_2 + O(s^3)$ is constrained, with $Q_1 = [A, Q_0]$ a purely off-diagonal Grassmann tangent, and the chiral generator is identified as the transported equivariance defect rate of Section 5,

$$[\Pi_{J_\Pi\text{-odd}}(1 - P)]_{LL} = \frac{1}{2}\Delta_\chi(P), \quad [\Pi_{J_\Pi\text{-odd}}\dot{Q}(0)]_{LL} = \frac{1}{2}\partial_s\Delta_\chi(P)|_0.$$

This reduces the second deliverable to a single target, the admissible normalisation of $\partial_s\Delta_\chi(P)|_0$, with $|u|$ remaining dictionary-bound. These are the last conceptual locks before the explicit construction of E_Π as a Lorentzian J_Π -odd projected Dirac residue.

12 Conclusion

The generation split u is now localised with no remaining ambiguity of method: it is controlled by the $\mathcal{D}^\pm$ -transported, generation-projected component of the antiunitary chiral equivariance defect

$$\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$$

of the spinorial directions eliminated by the non-injective projection, read off the Schur complement $E_\Pi = -M^\dagger M$ and restricted to the gauge-singlet triplet C_{gen}^3 . It vanishes identically for the J_Π -equivariant minimal non-injectivity, and is non-zero only through a chiral symmetry-breaking carried by the projection-locking axiom A4. The polarisation, the Seeley–DeWitt conventions, and the operator/spectral identification are locked; the front observables are confirmed to be null controls rather than carriers of u . The Schur-transverse branch is selected in the present Lorentzian spin stratum (Lemma 3, Corollary 3). Combined with the electric-genus result of the A4 companion [10], this promotes the existence statement to $u \neq 0$. It does not determine the value $|u|$, which remains fixed by the chiral-frontier normalisation through the Born–Infeld saturation structure [2].

A Numerical front controls

This appendix records the scalar front diagnostics obtained in the session runs of the oriented-frontier pipeline. They are the null controls referenced in Section 7: the scalar front explains the obstruction and bounds the maximal locking, but does not carry u , in accordance with Remark 2. The values below are the archived scalar-front null controls used in this note.

The oriented-frontier increment is the cocycle $\Delta A_c(g, s) = c(z(s) + a(g)s_b)$, reduced to $ca(g)s_b$ on admissible generators. Its symmetric front average vanishes identically,

$$\Theta_{\text{scalar}} = \max_{e \in \partial^+ S_m} |\theta_{\text{raw}}^+(e)| = 0 \quad (q \in \{61, 101, 151\}), \quad (18)$$

the signature of the φ -re-pairing of the oriented frontier. This is the null control, not a failure: it is the scalar shadow of the chiral asymmetry that lives in the Lorentzian residue (Section 7).

The maximal locking bound, normalised by the angle quantum $2\pi/q$, is q -invariant. The bare bound $\Theta_{\text{max}} = \langle |\Delta A_c| \rangle$ decays as $\sim 2.118/q$ (the $2\pi/q$ convention), so the q -invariant content is the product:

q	$\Theta_{\text{max}} \cdot q$
61	2.1141
101	2.1221
151	2.1183

giving $\Theta_{\text{max}} \cdot q \approx 2.118$, stable to within $\pm 0.2\%$ across the tested primes. This is the q -stable maximal-locking bound of the scalar front; it is a diagnostic of the bounded transport, not the spectral split u .

The pair-saturation and the BFS regime are

$$n_3^{\text{obs}} = 2, \quad |\text{nodes}| = 21569 \text{ at depth 14 (identical across } q), \quad (19)$$

confirming that the measurement lives at shell depth ≤ 2 on the q -independent nilpotent ball, in the pre-wraparound regime ($q^3 \gg 21569$). The saturation rank $n_3^{\text{obs}} = 2$ is the σ_{pair} saturation of the front and is distinct from the rank-three generation count $\sigma_c(n_3) = 3$ of [6], which is a separate admissible-rank observable. With only two contributing shells, the q -stability of (18)–(19) is the operative robustness check; the constancy of any per-shell profile is a weak test on two points.

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$[\Pi_{J_{\Pi\text{-odd}}}(1-P)]_{LL} = \frac{1}{2}\Delta_{\chi}(P)$, $[\Pi_{J_{\Pi\text{-odd}}}\dot{Q}(0)]_{LL} = \frac{1}{2}\partial_s\Delta_{\chi}(P)|_0$, reducing the value question to the admissible normalisation of $\partial_s\Delta_{\chi}(P)|_0$, doi:10.5281/zenodo.20763532.

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